QnVoice Urdu Text to Speech

An Offline Acoustic Model and Benchmark

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Abstract

QnVoice's Urdu speech candidate uses a three-step Matcha solver and a learned convolutional correction to produce a mel spectrogram. It retains the bundled Urdu encoder, duration predictor, and HiFi-GAN vocoder.

On an Apple M1 Pro, the student reduced median acoustic inference by 66% and median time for the full speech pipeline by 14% across 24 held-out prompts. Whisper word error rate was 19.3% for the student and 22.1% for the baseline. The small test set makes the difference in recognition scores inconclusive.

On an iPhone 15 Pro, median warm synthesis time was 1.30 s for the student and 1.50 s for the baseline.

Why Urdu Speech Runs Locally

Ethnologue's 2026 estimates put Urdu at about 246 million speakers worldwide [1]. Apple's iOS 27 Read & Speak language list omits Urdu [2]. QnVoice's device capability check also found no Urdu voice exposed to the app by iOS. A bundled acoustic model and vocoder give the app an offline way to speak Urdu text.

Scope

The input is Urdu script, including text entered directly or supplied by a translator. The output is mono speech at 22.05 kHz. The benchmark measures synthesis after text encoding. Translation quality, speech recognition during live conversations, and the time required to start app playback are separate from this experiment.

Example Input

مجھے ایک گلاس پانی چاہیے۔

I'd like a glass of water. Appendix A includes more test phrases and their recordings.

Acoustic Model

The teacher is the Apache 2.0 Matcha Urdu export by AhsanTalal, at the revision pinned in the QnVoice repository [3, 4]. Matcha applies conditional flow matching to construct speech features from noise and a duration-aligned representation of the text. The bundled graph performs ten Euler updates before the vocoder turns the resulting spectrogram into audio [7].

Table 1 Synthesis stages

Stage	Student configuration
Text encoder	Retained Urdu grapheme vocabulary, blank token interleaving, and duration prediction.
Acoustic solver	Retained flow estimator evaluated at times 0.0, 0.3, and 0.6 with step sizes 0.3, 0.3, and 0.4.
Mel correction	75,920 learned parameters in a residual network with three dilated convolution blocks.
Vocoder	Retained HiFi-GAN v1 export with 80 mel bands and 22.05 kHz mono output.

Training Procedure

We generated 384 unique Urdu prompts from 16 sentence templates covering places, names, objects, and times. A seeded split left 344 prompts for training and 40 for model selection. The training prompts correspond to about 15.9 minutes of teacher-generated speech by mel duration. Training and validation share template families. The evaluation uses 24 separately written prompts.

Each training example pairs the teacher’s ten-step mel with a three-step mel from the same text, using matching random seeds and the same predicted duration. We froze the retained graph and trained the correction for ten epochs with AdamW. The loss combines absolute mel error and a weighted error on changes between adjacent frames. The validation split selected the checkpoint.

The correction lowered validation mel mean absolute error from 0.1109 to 0.1082. We exported the trained layer into the acoustic graph and preserved its input and output names. The package records the seed, model hash, and evaluation prompts.

Evaluation

We synthesized 24 held-out phrases using the bundled ten-step teacher, the uncorrected three-step solver, and the trained student. For each variant we warmed the ONNX sessions, then timed three passes with two CPU threads using ONNX Runtime 1.24.2 on an Apple M1 Pro with 16 GB of memory. The benchmark rotated model order and paired noise seeds across variants. Pipeline time includes acoustic inference and waveform generation.

We ran Whisper small with Urdu selected on one generated clip per phrase per variant [8]. Word and character error rates compare its transcript with the input after normalizing punctuation and equivalent script forms. The evaluation set contains 181 words and 600 characters.

Mac Results

Table 2 Warm synthesis and recognition scores

Metric	Bundled Matcha	Three steps	Urdu student
Acoustic graph size	75.3 MB	73.9 MB	74.2 MB
Median acoustic time	274 ms	92 ms	94 ms
Median pipeline time	1.30 s	1.13 s	1.11 s
95th percentile pipeline time	1.81 s	1.50 s	1.43 s
Median real-time factor	0.49	0.42	0.41
Whisper word error rate	22.1%	19.9%	19.3%
Whisper character error rate	9.0%	7.2%	7.5%
Mel error versus teacher	—	0.1103	0.1078

Fresh Session Timing

On the glass-of-water sample, fresh acoustic and vocoder sessions plus synthesis took a median 5.15 s for Matcha and 1.79 s for the student. Each model ran five times in the same Mac process, so the operating system could reuse cached model files.

Interpretation

The student produced 35 word edits versus 40 for the teacher. The 95% paired bootstrap interval for the word error difference was -7.0 to +1.6 percentage points. The uncorrected solver had a slightly lower character error rate than the student. The learned layer primarily improved agreement with the teacher’s mel features. HiFi-GAN accounted for most of the elapsed pipeline time. Native Urdu listening tests remain necessary to assess intelligibility, pronunciation, and naturalness.

iPhone Results

Urdu Lab ran on an iPhone 15 Pro with iOS 27.0 using ONNX Runtime and two CPU threads. After warm-up, each model rendered the 24 held-out phrases three times. All 72 runs per model returned finite, non-silent waveforms. Timings exclude model loading and playback.

Table 3 Warm synthesis on iPhone 15 Pro

Model	Acoustic median	Pipeline median	Pipeline p95
Bundled Matcha	311 ms	1.50 s	2.08 s
Urdu student	103 ms	1.30 s	1.79 s

The longest synthesis time for the student was 1.99 s. Audio durations ranged from 2.07 to 3.36 s.

Model Package and Provenance

The GitHub companion project contains the 74.2 MB acoustic graph, the 55.8 MB vocoder, vocabulary, a command-line runner, paired WAVs, and benchmark records. The acoustic graph accepts x, x_lengths, and scales and returns mel and mel_lengths. GitHub companion project · Original Matcha export on Hugging Face

The teacher card identifies UrduSpeech as its training source [4, 5]. UrduSpeech identifies AI4Bharat's Rasa as its upstream source [5, 6]. Both public dataset cards list CC BY 4.0. This run uses synthetic teacher outputs, and the package carries the published attributions and modification notice. Commercial release review still needs the gated access terms and the scope of speaker rights alongside the published licenses [9].

References

[1] 2026 Ethnologue figures as reproduced by Visual Capitalist

[2] Apple iOS 27 feature availability

[3] Mehta and colleagues. Matcha-TTS. ICASSP 2024

[4] AhsanTalal Matcha Urdu TTS pinned export

[5] Humair Munir. UrduSpeech dataset card

[6] AI4Bharat. Rasa dataset and license

[7] Sherpa ONNX vocoder conversion release

[8] Radford and colleagues. Robust Speech Recognition via Large-Scale Weak Supervision

[9] Creative Commons Attribution 4.0 International

Appendix A Benchmark Samples

These six held-out prompts were used in both the Mac and iPhone benchmarks. The linked WAVs come from the first timed pass on the Mac and pair the original Matcha voice with the Urdu student. The files are in samples/benchmark in the GitHub companion project.

Table 4 Test phrases and paired recordings

ID	Urdu prompt	English translation	Evaluation audio
02	مجھے ایک گلاس پانی چاہیے۔	I'd like a glass of water.	02-baseline.wav 02-student.wav
03	ذرا دائیں طرف مڑیں اور پل تک چلتے رہیں۔	Please turn right and keep going until you reach the bridge.	03-baseline.wav 03-student.wav
12	آج ٹرین دس منٹ تاخیر سے آئے گی۔	The train will arrive ten minutes late today.	12-baseline.wav 12-student.wav
15	مجھے سمجھ نہیں آئی، آہستہ بولیے۔	I didn't understand. Please speak slowly.	15-baseline.wav 15-student.wav
18	واپسی کا ٹکٹ کہاں سے ملے گا؟	Where can I get a return ticket?	18-baseline.wav 18-student.wav
22	کل کی ملاقات دو بجے ہوگی۔	Tomorrow's meeting will be at two o'clock.	22-baseline.wav 22-student.wav